



Aditya University

School Computing
Department of Computer Applications

IoT PROJECT – REVIEW-I

Project: IoT-Based Smart

Traffic Management System

Programme: MCA Year: 2 Semester: III

Review-I: 24-08-2026 to 28-08-2026

Student Team:

- 1. Ch.vinay(25M11MC168)**
- 2. G.Bharath(25M11MC048)**
- 3.M.Rahul(25M11MC090)**
- 4.Bindu Sagar Patro(25M11MC014)**



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IoT-Based Smart Traffic Management System

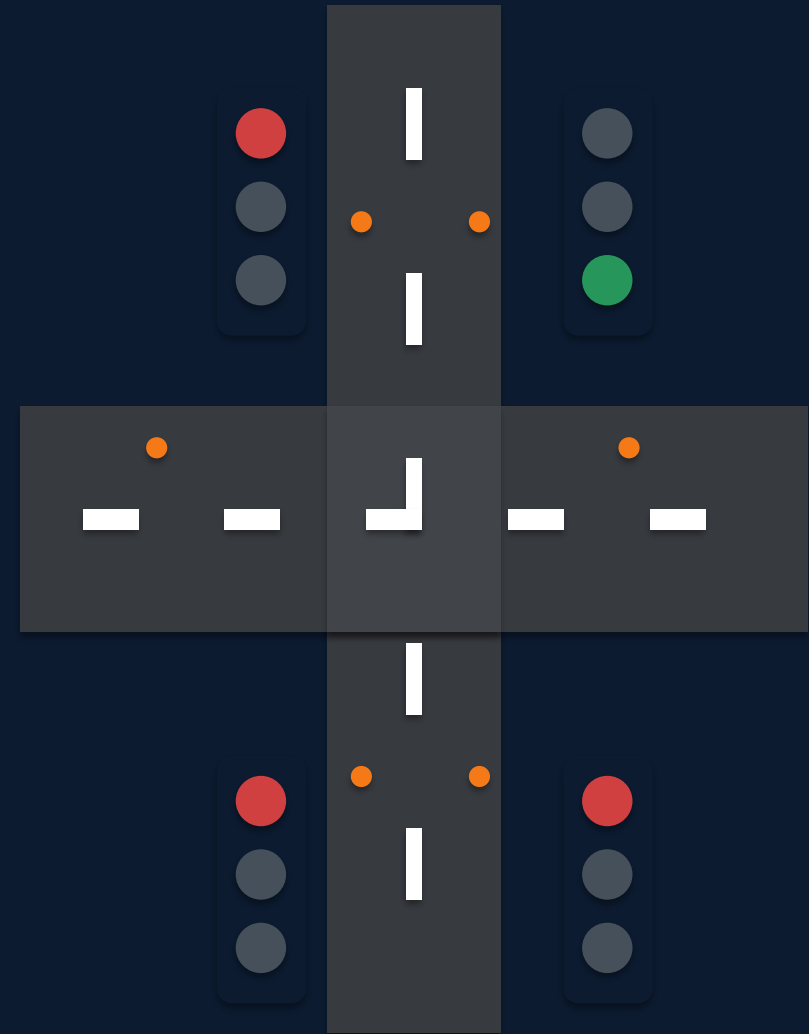
REVIEW – I | MCA | Smart & Adaptive Traffic Control

CORE IDEA

Detect traffic → Analyze density → Control signals → Monitor through IoT

PROJECT GOAL

Reduce unnecessary waiting through traffic-aware signal timing.



1. Project Abstract

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A concise overview of the proposed system

- Monitors traffic density on different roads in real time.
- Uses sensors and ESP32 to process road-wise traffic information.
- Selects the road with higher traffic for signal priority.
- Adjusts green time according to the detected traffic level.
- Uses IoT/Wi-Fi to display traffic status remotely.

KEY IDEA

More Traffic → More Green Time

CONTROLLER

ESP32 = Processing + Wi-Fi

EXPECTED BENEFIT

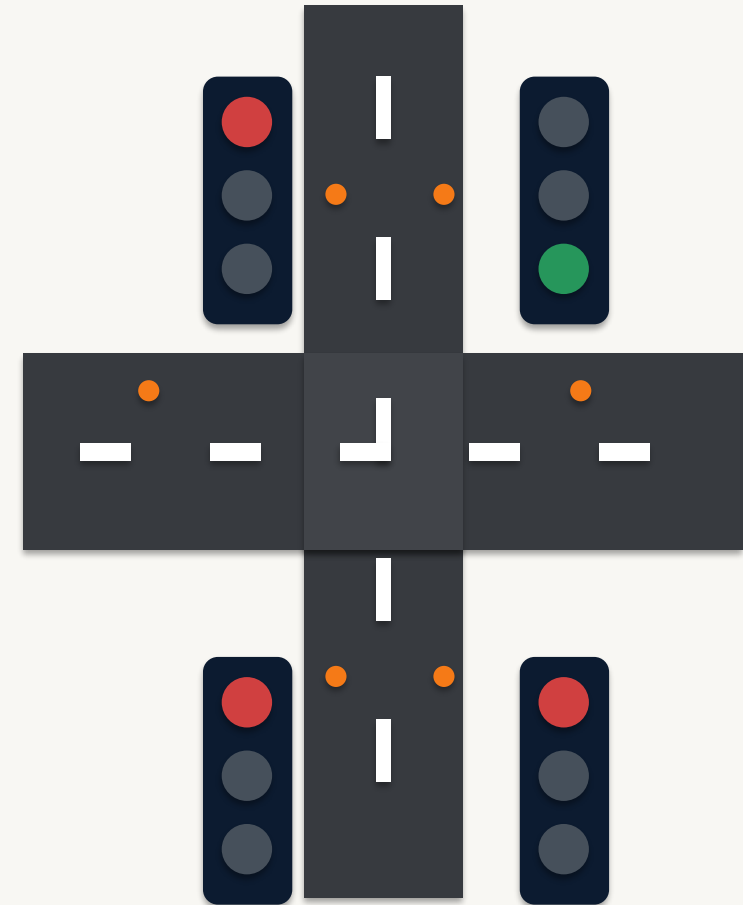
Less unnecessary waiting

2. Introduction

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Why a traffic signal should respond to real traffic

- Urban traffic varies continuously across roads and time.
- Traditional signals commonly use predefined or fixed timing.
- Fixed timing cannot fully respond when one road becomes heavily congested.
- Real-time sensing provides the information required for adaptive control.
- IoT enables remote visibility of the current traffic condition.



SHIFT

Fixed Timing → Traffic-Aware Timing

3. Problem Statement

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Limitations of the existing fixed-time approach

EXISTING PROBLEM

- Fixed green-light duration
- Unequal traffic on different roads
- Long queues may continue waiting
- Unnecessary vehicle idling
- Limited real-time monitoring

OUR PROPOSED DIRECTION

- Detect actual traffic density
- Identify the priority road
- Adjust green time dynamically
- Monitor status through IoT
- Recalculate after each cycle

4. Project Objectives

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The main outcomes expected from the prototype

DETECT

Vehicle presence / density

ANALYZE

Classify Low / Medium / High

PRIORITIZE

Identify the highest-demand road

CONTROL

Adjust green-light duration

MONITOR

Display live IoT status

IMPROVE

Reduce waiting and congestion

Detect → Analyze → Prioritize → Control → Monitor

5. Proposed IoT Modules / Functionalities

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Six core modules for the prototype

01 VEHICLE DETECTION

Sensors detect vehicle presence.

02 DENSITY ANALYSIS

ESP32 classifies traffic level.

03 SMART SIGNAL CONTROL

Higher traffic gets priority green time.

04 IoT COMMUNICATION

Wi-Fi transmits the latest status.

05 DASHBOARD

Displays road-wise traffic and active signal.

06 ALERTS

Highlights high-congestion conditions.

SENSORS

ESP32

DECISION

SIGNAL

IoT

6. Hardware Requirements & Software Requirements

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Prototype components and development tools

HARDWARE

- ESP32 — main controller + Wi-Fi
- IR Sensors — vehicle detection
- Red / Yellow / Green LEDs — signals
- Resistors — LED protection
- Breadboard & jumper wires — prototype
- Optional OLED / buzzer — display & alert

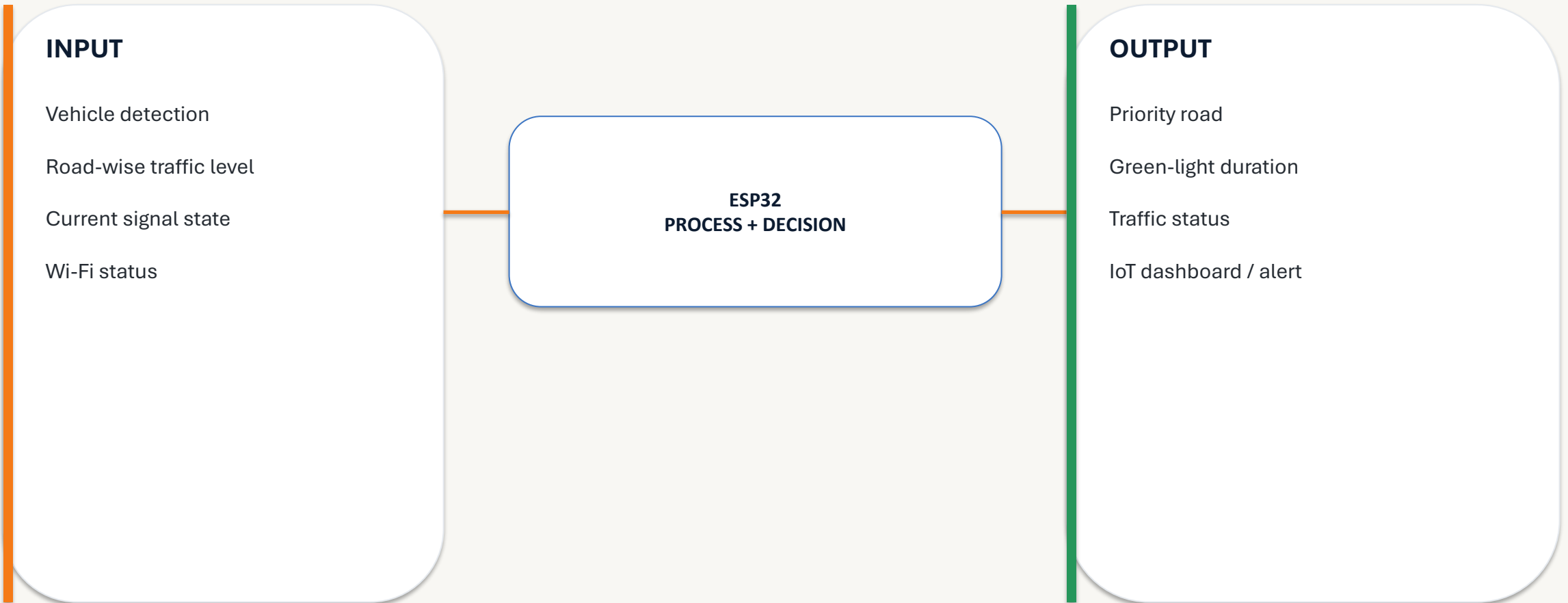
SOFTWARE

- Arduino IDE
- Embedded C/C++
- ESP32 Wi-Fi libraries
- IoT / Cloud platform
- Dashboard interface
- Serial Monitor for testing

7. Input and Expected Output

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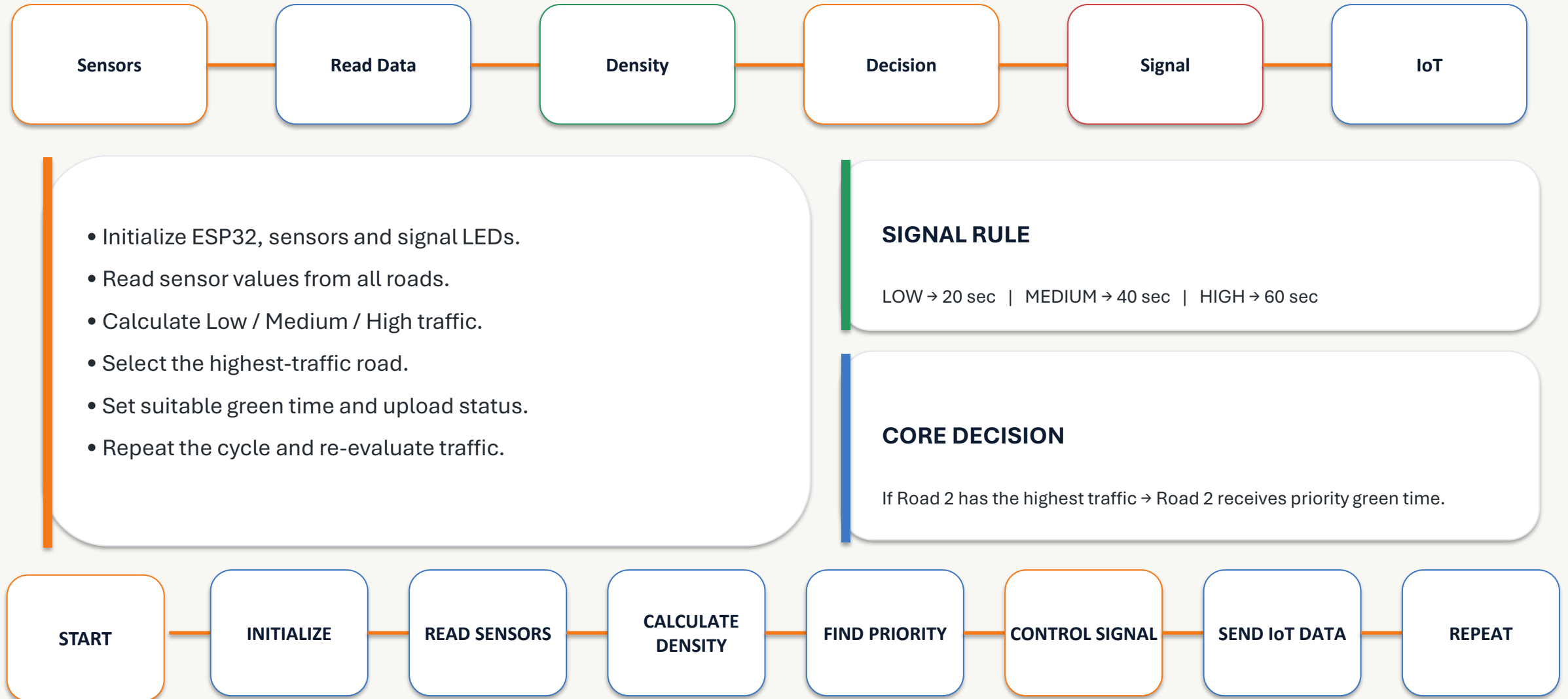
How sensor data becomes a traffic-control decision



8. Design Methodology: Algorithm & System Flow

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Simple logic that can be demonstrated clearly



9. Implementation Progress

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Current status for Review-I

Project analysis & objectives

COMPLETED

Hardware / software selection

COMPLETED

Architecture & flowchart

COMPLETED

Sensor interfacing

IN PROGRESS

ESP32 programming

IN PROGRESS

Smart signal logic

IN PROGRESS

IoT dashboard

PLANNED

Complete prototype testing

PLANNED

REVIEW-I FOCUS

Complete sensor + ESP32 + adaptive signal integration before IoT/dashboard validation.

10. Completed Module Demonstration & Sample Output

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The most important live-demo slide

LIVE DEMO SEQUENCE

1. Put more toy cars on Road 2.
2. Sensor detects higher traffic.
3. ESP32 calculates density.
4. Road 2 gets longer green time.
5. Move cars to Road 1.
6. Priority changes to Road 1.

SMART TRAFFIC DASHBOARD

Road 1

LOW

Road 2

HIGH

Road 3

MEDIUM

Road 4

LOW

CURRENT DECISION

GREEN ROAD : 2
GREEN TIME : 60 sec
STATUS : HIGH TRAFFIC

11. Individual Team Member Contribution

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Clear ownership of technical tasks

MEMBER 1 — Analysis & Documentation

Problem, literature, report & presentation

MEMBER 2 — Hardware & Sensors

Components, sensor circuit & hardware testing

MEMBER 3 — ESP32 & Algorithm

Programming, density & signal-control logic

MEMBER 4 — IoT & Testing

Wi-Fi, dashboard & integration testing

TEAM PRINCIPLE

Every member understands the complete system, even if tasks are divided.

12. Challenges Identified

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Main technical risks in the prototype

SENSOR ACCURACY

- False or inconsistent detection
- Sensor placement
- Vehicle position / distance

ALGORITHM

- Choosing density thresholds
- Fair signal timing
- Changing traffic conditions

IoT & INTEGRATION

- Wi-Fi reliability
- Wiring issues
- Real-time dashboard updates

13. Proposed Solutions

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Practical steps to improve reliability

- Calibrate sensors using repeated tests at different positions.
- Use clear Low / Medium / High thresholds and tune them experimentally.
- Set minimum and maximum green times to maintain fairness.
- Add Wi-Fi reconnection so local control can continue during network loss.
- Test every module separately before complete integration.

SENSOR

Calibrate + repeat tests

ALGORITHM

Tune thresholds + timing

INTEGRATION

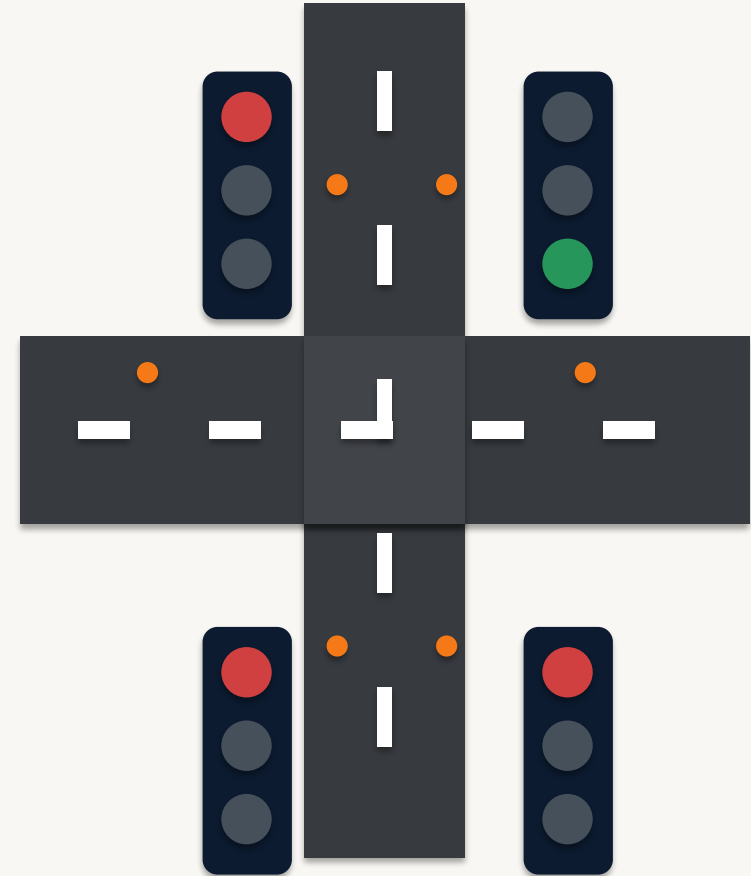
Module testing → full testing

14. Conclusion

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Final message for Review-I

- Detects traffic conditions using sensors.
- Uses ESP32 for traffic analysis and signal decisions.
- Provides adaptive signal priority based on traffic density.
- Uses IoT for real-time remote monitoring.
- Creates a foundation for future emergency-vehicle priority and advanced analytics.

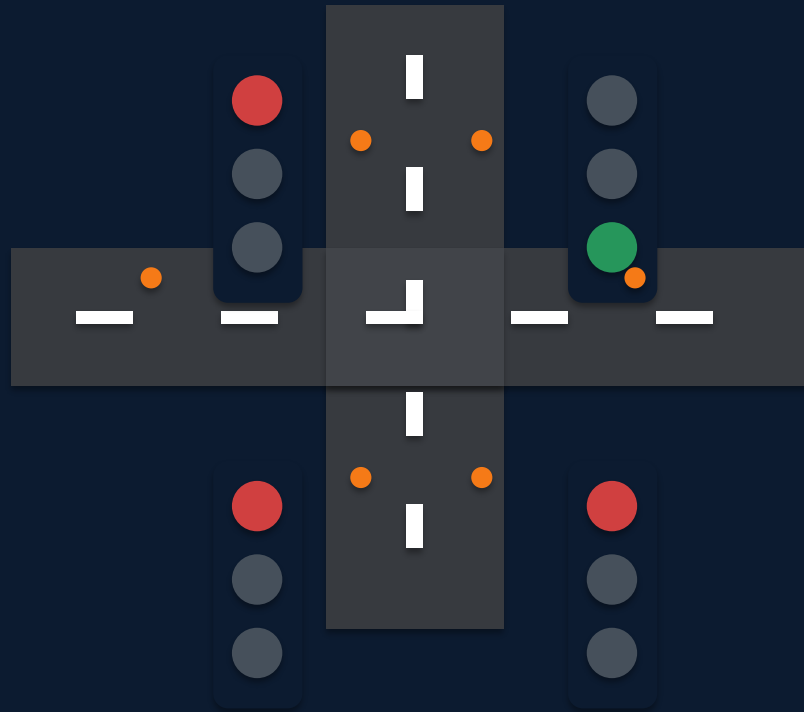


TAKEAWAY

Smart Traffic = Real-Time + Adaptive + Connected

THANK YOU

Questions & Suggestions



DETECT • ANALYZE • CONTROL • CONNECT